Chapter 10

Consumption Taxes

This chapter covers consumption taxes, which can be divided into two board categories:

1. General sales taxes: Those taxes cover all consumption goods with few exceptions (e.g., food). In the United States, retail sales taxes are widely used, whereas in the European Union, the value-added tax is more common.
2. Excise taxes: Those taxes apply to specific goods and can take various forms depending on the purpose of the tax.

We will cover the definition of the tax base as well as the applicable tax rates. An example of tax rates can be found through resources like the Federation of Tax Administrators: Tax Rates. Finally, the evaluation of sales taxes involves assessing their effectiveness and implications in the broader economic context.

10.1 Consumption Tax Overview

The theoretical relationship between consumption, income, and savings is straightforward: Consumption equals individual income plus the change in net wealth. Recall that this is the Haig-Simons definition from the previous chapter. An ideal (theoretical) consumption tax would be broad and applied to all consumption. However, such a tax would be challenging to identify, collect, and enforce, making it impractical from a financial administration perspective. In reality, consumption taxes tend to be narrower in scope as they are typically limited to certain goods or services. These taxes can be collected at various stages of production, including the resource supply stage (such as mining or farming), during manufacturing, at the wholesale level, or at the point of retail.

There are various types of consumption taxes, each differing in how and when they are applied throughout the production process. A single-stage tax is applied at one point in the process. An example is the retail sales tax, which is imposed

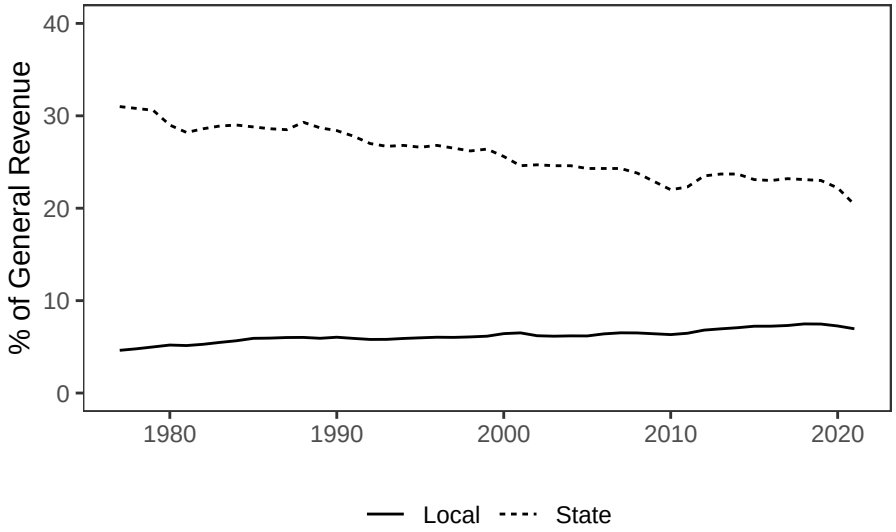


Figure 10.1: Sales Tax Revenue at State and Local Level. Source: Tax Policy Center State General Revenue, by Source, Percentage Distribution, 1977-2021.

at the point of sale to consumers. A multi-stage tax applies to more than one point in the production process. The turnover or gross receipts tax is an example where a tax is levied on sales at each stage, leading to cascading effects as the tax accumulates through the production chain. The value-added tax is also a multi-stage tax but is non-cascading, i.e., the tax is based on the value added at each stage, preventing the overall tax burden from increasing with each transaction. Excise taxes are imposed on a single commodity and are generally single-stage taxes. They apply specifically to one type of product, such as gasoline or tobacco, and are collected at one point in the production or distribution process.

10.2 Retail Sales Taxes and Value-Added Taxes

A general sales tax serves as an alternative to income and other taxes to finance government operations. The broad base is one of the advantages because it allows for high revenue generation at a low tax rate. A general sales tax implicitly taxes wealth because wealthier people consume more and promotes economic growth by discouraging consumption and encouraging savings.

The collection of a general sales tax can take different forms. A retail sales tax is imposed only at the final stage of production or distribution. It is commonly used by U.S. state and local governments, as well as in Canadian provinces (Figure 10.1). On the other hand, the value-added tax applies to every transaction

along the production and distribution chain. Each purchaser, except for the final consumer, can claim a refund on the tax through credits. The value-added tax is a requirement for membership in the European Union and is also used in countries like China, Russia, and several in Latin America. A value-added tax is self-enforcing (more details about this later), which leads to the ability to impose higher tax rates. A useful resource is available from the Tax Policy Center on why the VAT is considered administratively superior to a retail sales tax.

10.2.1 Retail Sales Tax

In the U.S., retail sales taxes typically exclude goods used in production for reasons related to economic efficiency, equity, and the promotion of economic growth. The exclusion is meant to prevent the tax from being applied to intermediate goods, which could otherwise lead to a cascading effect and increase production costs. The standards for these exclusions usually involve two criteria: (1) The good must either be a component or physical ingredient of the final product or (2) it must be directly used in the production process.

Sales taxes in the U.S. are often composed of state, local, and special rates. For example, in New York City, the total sales tax is a combination of a 4% state tax, a 4.5% city tax, and a 0.375% surcharge for the Metropolitan Commuter Transportation District (see New York City Department of Finance for more information). Resources like the Tax Foundation's data portal provide detailed information on state and local sales tax rates across the country. Note that sales taxes are not levied in Alaska, Delaware, Montana, New Hampshire, and Oregon.

Retail sales tax can be either general or selective. A general tax applies to all transactions except for certain exemptions, whereas a selective tax targets only specific, enumerated transactions, such as a lodging tax that applies only to hotel stays. The way rates are applied can also differ with two primary types: Specific and ad-valorem. A specific, or unit tax, is applied based on the number of physical units bought or sold, such as a tax of $0.20 per gallon of gasoline. In contrast, an ad-valorem tax is levied based on the value of the transaction. For example, a lodging tax might be 10% of the total hotel bill, meaning the tax amount fluctuates according to the price paid for the service.

10.2.2 Value-Added Tax (VAT)

The mechanism of a value-added tax (VAT) involves applying the tax to the value added at each stage of production and distribution. Businesses charge VAT on their sales and subtract the VAT they have paid on their purchases when remitting the tax to the government. This system ensures that taxes are collected only on the added value at each stage, thereby eliminating the issue of cascading taxes that can occur when businesses pay sales tax on items they use in their operations, such as equipment, which they do not resell. There are three

	Farmer	Miller	Baker	Total tax
Prices charged				
No tax	300	700	1000	0
Sales tax	300	700	1100	100
VAT	330	770	1100	
Credit invoice method				
Tax liability	30	70	100	200
Tax credit	0	30	70	100
Subtraction method				
Tax liability	30	40	30	100

Table 10.1: Credit invoice vs. subtraction method to calculate the value-added tax.

types of VAT, each differing in how the tax is calculated (Table 10.1). Note that the examples are taken from the Tax Policy Center: How would a VAT be collected?.

- **Credit invoice method:** Each business charges VAT on everything it sells and records it on the sales invoice. It then subtracts the VAT it already paid on its purchases, as shown on its suppliers' invoices, and remits the difference. If it paid more VAT than it collected, it gets a refund. Example: A miller sells flour for 150 and bought wheat for 100. At 5%, it collects 7.50 from the baker, claims a credit of 5 for the VAT on the wheat, and remits 2.50. The method relies on two records: sales invoices and purchase invoices.
- **Subtraction Method:** The business skips the invoice-by-invoice credits. It takes total sales for the period, subtracts total purchases from other businesses, and applies the tax rate to the difference, which is its value added. Example: The miller's value added is 150-100=50, and 5% of 50 is 2.50, the same as under the credit invoice method.
- **Addition Method:*** Instead of starting from sales and purchases, the business adds up what it pays to the factors of production: Wages, interest, rent, and profit. That sum is also its value added, and the tax rate is applied to it. For example, a business with sales of $200,000, materials costing $125,000, and wages totaling $50,000 would use the addition method to calculate VAT. The VAT would be applied to $75,000, which is the difference between sales and material costs ($200,000-$125,000). Alternatively, under another interpretation of the addition method, the VAT would be calculated on $75,000, which consists of $50,000 in wages and $25,000 in profits.

The VAT offers several advantages, particularly in terms of reducing tax evasion and improving compliance. It is considered a self-enforcing tax because businesses are required to document receipts at every stage of production. This creates a paper trail that ensures taxes are paid at each step, making it difficult to

			Tax Type			
Stage	Sales	Value added	Sales	VAT	Turnover	Manufacturer
Tax rates			5%	5%	2.11%	4.44%
Farmer	100	100		5.00	2.11	
Miller	150	50		2.50	3.16	6.67
Baker	300	150		7.50	6.32	13.33
Retailer	400	100	20.00	5.00	8.42	
Total tax			20.00	20.00	20.00	20.00

Table 10.2: Comparison of revenue generation from various types of consumption taxes. Note that all tax types result in the same amount of government revenue.

evade the tax. In contrast, retail sales taxes often provide more incentive for cheating, particularly when rates exceed 10%, as implementation tends to be less successful. Another advantage of the VAT is that it helps level the playing field between foreign and domestic products. Some countries choose to exempt domestic taxes from goods sold internationally, ensuring that exported goods are not unfairly taxed when competing in global markets. This approach can encourage international trade by removing the burden of domestic taxes on exports. The various general sales taxes can be designed in such a way to all raise the same amount of tax revenue (Table 10.2).

10.3 Excise Taxes

Excise taxes are taxes imposed on specific goods, services, or activities rather than on consumption generally. Unlike broad-based consumption taxes, such as retail sales taxes and VAT, excise taxes selectively target particular products. They may be imposed as a fixed amount per unit sold (specific excise taxes) or as a percentage of the product's value (ad valorem excise taxes). Examples include taxes on gasoline, tobacco, alcoholic beverages, and certain luxury goods.

Governments impose excise taxes for several reasons, including raising revenue, discouraging consumption, correcting negative externalities, financing public services, and redistributing tax burdens. These objectives are not mutually exclusive. For example, gasoline taxes generate government revenue, help finance transportation infrastructure, and can discourage fuel consumption and associated emissions. However, the appropriate design of an excise tax depends on its primary policy objective. For an overview of federal excise taxes, see the Tax Policy Center: What are the major federal excise taxes, and how much money do they raise?.

One of the principal objectives of excise taxation is to generate government revenue. Excise taxes are frequently imposed on goods with relatively inelastic

demand, such as gasoline, tobacco, and alcoholic beverages. When demand is inelastic, an increase in the tax-inclusive price results in a relatively small reduction in quantity demanded. Consequently, these goods can provide a relatively stable source of tax revenue. The revenue generated by a specific excise tax can be expressed as $R = t \cdot Q(t)$ where R represents government revenue, t is the tax per unit, and $Q(t)$ is the quantity consumed at that tax rate. An increase in the tax rate raises revenue per unit but may also reduce consumption. Therefore, the relationship between the tax rate and total revenue depends on the responsiveness of demand and supply to the tax.

Excise taxes may also be used to pursue distributional objectives. For example, governments have occasionally imposed additional taxes on luxury goods, such as expensive automobiles, yachts, jewelry, and private aircraft, with the intention of placing a greater tax burden on high income households. However, the effectiveness of luxury taxes as redistributive instruments depends on consumer behavior, the availability of substitutes, and the economic incidence of the tax. An example is New Jersey's Luxury and Fuel-Inefficient Vehicle Surcharge, which imposes a one-time surcharge of 0.4% on qualifying new passenger vehicles with a sales or lease price of at least $45,000 or a fuel efficiency rating below 19 miles per gallon. This tax combines elements of luxury taxation and environmental taxation. At the federal level, the Omnibus Budget Reconciliation Act of 1990 introduced luxury excise taxes on selected products, including expensive automobiles, yachts, jewelry, furs, and private aircraft. These taxes were subsequently repealed or allowed to expire.

Finally, governments may use selective excise taxes to shift part of the tax burden to nonresidents. For example, taxes on goods or services disproportionately purchased by tourists may generate revenue from visitors rather than local residents. The extent to which this occurs depends on economic incidence and the responsiveness of tourists and businesses to taxation.

Let us next outline three types of excise taxes: (1) Sumptuary taxes, (2) benefit-based taxes, and (3) regulatory and environmental taxes.

Sumptuary Excise Taxes (Sin Taxes): Sumptuary taxes, which are also known as sin taxes, are imposed on goods whose consumption governments seek to discourage because of concerns about health, safety, or other social consequences. The most common examples are excise taxes on tobacco and alcoholic beverages (see for example How Hard Do Distilled Spirits Taxes Bite in Your State?). Sin taxes generally serve two objectives: raising government revenue and reducing consumption. The effectiveness of these taxes in discouraging consumption depends on the price elasticity of demand. When demand is relatively inelastic, higher taxes may generate substantial revenue but lead to relatively small reductions in consumption. When demand is more elastic, the same tax increase may produce a larger reduction in consumption. Importantly, demand elasticities may differ across consumer groups. For example, younger consumers may respond differently to cigarette price increases than established adult smokers (see for example To Reduce the Cost of Teenage Temptation, Why Not Just Raise the

Price of Sin?). Consequently, the effects of sin taxes on consumption, government revenue, and household welfare may vary across the population. Although sin taxes are often justified by the adverse consequences of consumption, it is useful to distinguish between costs borne by consumers themselves and external costs imposed on others. The latter provide a conventional economic justification for corrective taxation. Taxes intended to discourage behavior considered harmful to the consumer may involve additional policy considerations, including imperfect information, addiction, and self-control problems.

Benefit-Based Excise Taxes: Excise taxes may also be used to finance public services that are closely related to the taxed activity. Under the benefit principle of taxation, individuals who benefit from government-provided services should contribute toward their financing. A common example is the taxation of motor fuels to finance transportation infrastructure. Federal gasoline and diesel excise taxes provide revenue for the Highway Trust Fund, which supports federal highway programs and certain public transportation expenditures. Fuel taxes approximate a user charge because individuals who drive more generally purchase more fuel and therefore pay more in fuel taxes. However, the relationship is imperfect. Fuel consumption depends not only on distance traveled but also on vehicle fuel efficiency. Consequently, two drivers traveling the same distance may contribute different amounts toward transportation infrastructure. Fuel taxes also have administrative advantages. Collecting taxes from fuel producers and distributors can be less costly than collecting small payments from individual road users. However, fuel taxes do not directly reflect the specific costs imposed by individual drivers, such as road congestion, pavement damage, or infrastructure use at particular locations and times. Another limitation is that fuel excise taxes are generally imposed as a fixed monetary amount per gallon. Unless tax rates are periodically adjusted or indexed to inflation, their real value declines over time. In addition, improvements in vehicle fuel efficiency and the adoption of electric vehicles can weaken the relationship between road use and fuel-tax revenue. These considerations have motivated interest in alternative transportation funding mechanisms, including vehicle-miles-traveled charges and congestion pricing.

Regulatory and Environmental Excise Taxes: Regulatory and environmental excise taxes are designed to influence economic behavior by increasing the private cost of activities that impose costs on society. When production or consumption generates negative externalities, market prices may fail to reflect the full social costs of an activity. By imposing a tax, governments can encourage consumers and producers to account for some or all of these external costs. This is the economic rationale for Pigouvian taxation, which was introduced in the chapter on the role of government in the economy. Under the standard Pigouvian framework, the corrective tax is set equal to the marginal external damage at the socially efficient level of activity. Environmental excise taxes may be imposed on fossil fuels, emissions, or products associated with environmental damages. For example, the federal gas guzzler tax applies to certain new passenger cars that fail to meet specified fuel-economy standards. The tax is intended to discourage

the purchase of relatively fuel-inefficient vehicles. Unlike benefit-based taxes, which are principally intended to finance public services, corrective taxes are designed to influence behavior by incorporating external costs into private decisions. Nevertheless, a single excise tax may serve both purposes. For example, gasoline taxes can generate transportation revenue while also discouraging fuel consumption. However, a fuel tax based on gallons purchased is not necessarily equivalent to a tax designed to address specific externalities, such as traffic congestion or vehicle emissions. Different externalities may require different tax bases or tax rates. The effectiveness of environmental excise taxes depends on how closely the tax corresponds to the external damage and how consumers and producers respond to changes in prices. When direct measurement of an externality is difficult or administratively costly, governments may tax a related activity as an imperfect proxy.